



organic chemistry

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

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Time:

Marks: **73 marks**

Comments:

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**1.**

This question is about polymers.

- (a) Polyesters are produced when monomers join together and lose a small molecule.

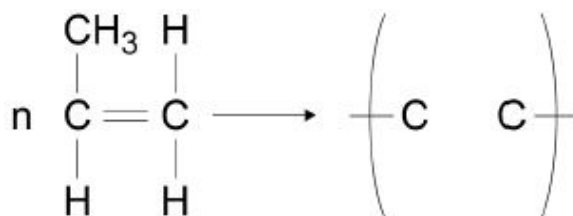
Name the small molecule lost.

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(1)

- (b) Poly(propene) is produced from propene.

Complete the structure of poly(propene) in the equation.



(3)

- (c) Carpets are made from:

- poly(propene)
- wool
- a mixture of poly(propene) and wool.

Poly(propene) wears out more slowly than wool.

A mixture of poly(propene) and wool to make carpets is more sustainable than using just poly(propene) or just wool.

Suggest why.

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(2)

Polymer fibres are used to make firefighter uniforms.

The table below shows some properties of two polymer fibres.

|                              | Polymer fibres |           |
|------------------------------|----------------|-----------|
| Property                     | Poly(propene)  | Polyester |
| Density in g/cm <sup>3</sup> | 0.90           | 1.38      |
| Melting point in °C          | 165            | 260       |
| Flame resistance             | Poor           | Good      |
| Water absorption             | Low            | High      |

- (d) Evaluate the suitability of poly(propene) and polyester for firefighter uniforms.

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(4)

(Total 10 marks)

2.

This question is about combustion of fuels.

- (a) Some central heating boilers use wood as a fuel.

Suggest **two** reasons why wood is more sustainable than natural gas as a fuel for central heating boilers.

1 \_\_\_\_\_

\_\_\_\_\_

2 \_\_\_\_\_

\_\_\_\_\_

(2)

Natural gas is mainly methane.

When methane burns it can produce both carbon monoxide and carbon dioxide.

- (b) Explain the process by which carbon monoxide can be produced when methane is burned.

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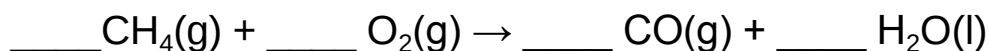
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(2)

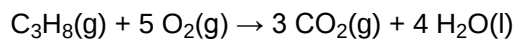
- (c) Balance the equation for the combustion of methane to produce carbon monoxide.



(1)

- (d) Propane burns to form carbon dioxide and water.

The equation for the reaction is:



3.60 dm<sup>3</sup> carbon dioxide is produced when a sample of propane is burned in 7.25 dm<sup>3</sup> oxygen.

Calculate the volume of unreacted oxygen.

Give your answer in cm<sup>3</sup>

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Volume of unreacted oxygen = \_\_\_\_\_ cm<sup>3</sup>

(4)

(Total 9 marks)

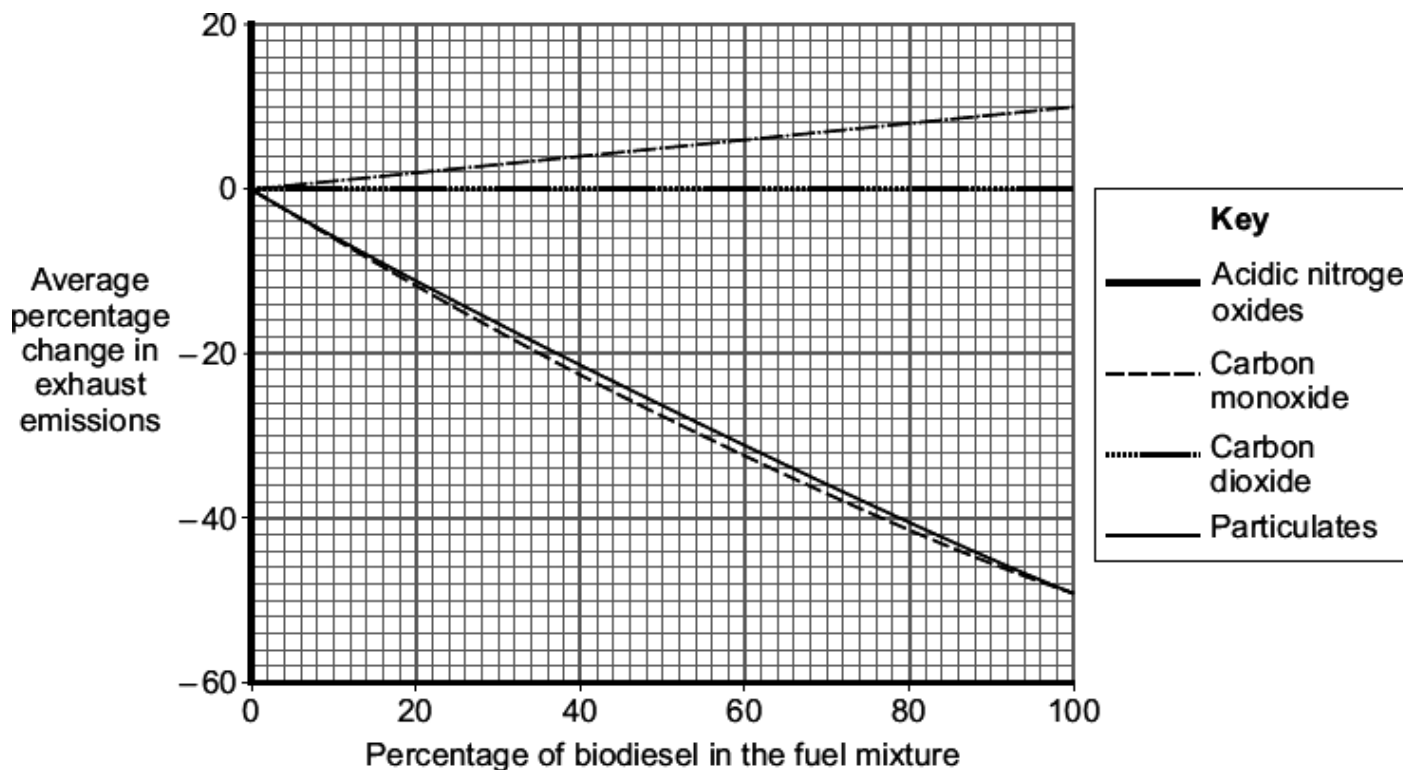
3.

Petroleum diesel is produced from crude oil.

Most vehicles that use petroleum diesel as fuel can also use biodiesel or a mixture of these two fuels. In the UK (in 2010) there must be 5 % biodiesel in all petroleum diesel fuel.

Biodiesel is produced from plant oils such as soya. The crops used to produce biodiesel can also be used to feed humans. The benefit that biodiesel is 'carbon neutral' is outweighed by the increasing demand for crops. This increasing demand is causing forests to be burnt to provide land for crops to produce biodiesel. Only a huge fall in the price of petroleum diesel would halt the increasing use of biodiesel.

The graph shows the average percentage change in exhaust emissions from vehicles using different mixtures of petroleum diesel and biodiesel.



There is no difference in carbon dioxide emissions for all mixtures of petroleum diesel and biodiesel.

Use the information and your knowledge and understanding to evaluate the use of plant oils to produce biodiesel.

Remember to give a conclusion to your evaluation.

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(5)

(Total 5 marks)

4.

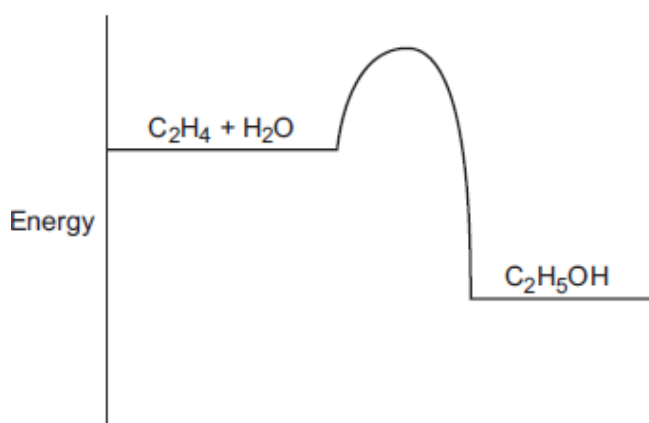
This question is about ethanol.

(a) Ethanol is produced by the reaction of ethene and steam:



(i) **Figure 1** shows the energy level diagram for the reaction.

**Figure 1**



How does the energy level diagram show that the reaction is exothermic?

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(1)

- (ii) A catalyst is used for the reaction.

Explain how a catalyst increases the rate of the reaction.

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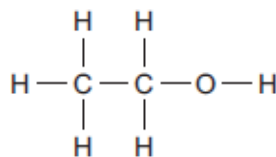
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(2)

- (b) **Figure 2** shows the displayed structure of ethanol.

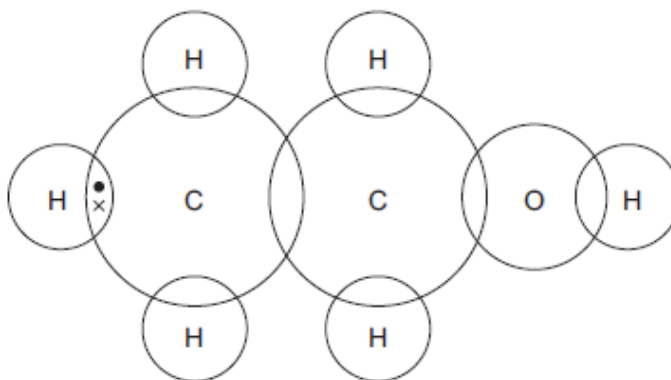
**Figure 2**



Complete the dot and cross diagram in **Figure 3** to show the bonding in ethanol.

Show the outer shell electrons only.

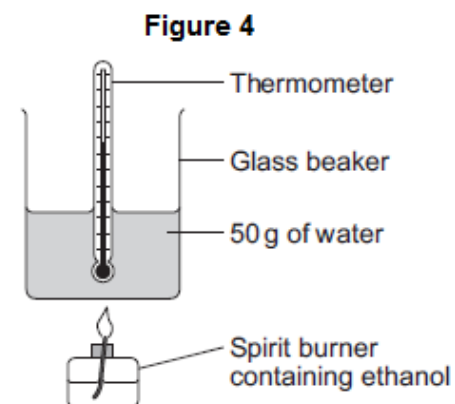
**Figure 3**



(2)

- (c) A student burned some ethanol.

**Figure 4** shows the apparatus the student used.



- (i) The student recorded the temperature of the water before and after heating.

His results are shown in **Table 1**.

**Table 1**

|                            |         |
|----------------------------|---------|
| Temperature before heating | 20.7 °C |
| Temperature after heating  | 35.1 °C |

Calculate the energy used to heat the water.

Use the equation  $Q = m \times c \times \Delta T$

The specific heat capacity of water = 4.2 J / g / °C

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Energy used = \_\_\_\_\_ J

**(3)**



- (ii) **Table 2** shows the mass of the spirit burner before the ethanol was burned and after the ethanol was burned.

**Table 2**

|                                                 |         |
|-------------------------------------------------|---------|
| Mass of spirit burner before ethanol was burned | 72.80 g |
| Mass of spirit burner after ethanol was burned  | 72.10 g |

Calculate the number of moles of ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ) that were burned.

Relative atomic masses ( $A_r$ ): H = 1; C = 12; O = 16

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Number of moles burned = \_\_\_\_\_

**(3)**

- (iii) Calculate the energy released in joules per mole.

You should assume that all the energy from the ethanol burning was used to heat the water.

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Energy = \_\_\_\_\_ J / mole

**(1)**

- (d) The names, structures and boiling points of ethanol and two other alcohols are shown in **Table 3**.

**Table 3**

| Name                | Methanol                                                                                             | Ethanol                                                                                                                                                     | Propanol                                                                                                                                                                                                           |
|---------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Structure           | $\begin{array}{c} \text{H} \\   \\ \text{H}-\text{C}-\text{O}-\text{H} \\   \\ \text{H} \end{array}$ | $\begin{array}{c} \text{H} \quad \text{H} \\   \quad   \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\   \quad   \\ \text{H} \quad \text{H} \end{array}$ | $\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\   \quad   \quad   \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\   \quad   \quad   \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$ |
| Boiling point in °C | 65                                                                                                   | 78                                                                                                                                                          | 97                                                                                                                                                                                                                 |

Use your knowledge of structure and bonding to suggest why the boiling points increase as the number of carbon atoms increases.

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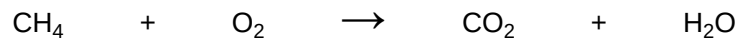
(3)

(Total 15 marks)

**5.**

This question is about energy changes in chemical reactions.

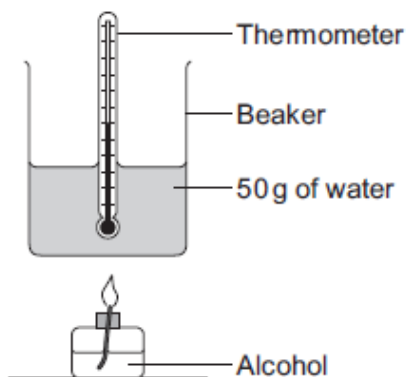
- (a) Balance the chemical equation for the combustion of methane.



(1)

- (b) Alcohols are used as fuels.

A group of students investigated the amount of energy released when an alcohol was burned. The students used the apparatus shown in the diagram below.



In one experiment the temperature of 50 g of water increased from 22.0 °C to 38.4 °C. The mass of alcohol burned was 0.8 g.

Calculate the heat energy (Q) in joules, released by burning 0.8 g of the alcohol. Use the equation:

$$Q = m \times c \times \Delta T$$

Specific heat capacity (c) = 4.2 J / g / °C

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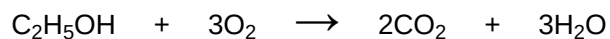
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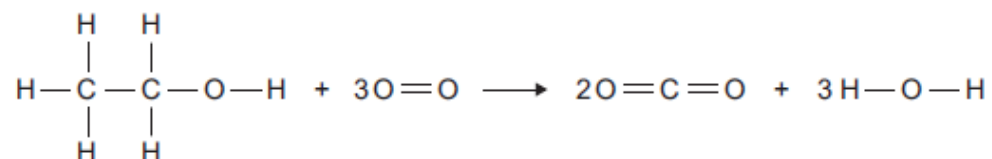
Heat energy (Q) = \_\_\_\_\_ J

(3)

- (c) The chemical equation for the combustion of ethanol is:



- (i) The equation for the reaction can be shown as:



| Bond  | Bond energy in kJ per mole |
|-------|----------------------------|
| C — H | 413                        |
| C — C | 347                        |
| C — O | 358                        |
| C = O | 799                        |
| O — H | 467                        |
| O = O | 495                        |

Use the bond energies to calculate the overall energy change for this reaction.

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Overall energy change = \_\_\_\_\_ kJ per mole

(3)

- (ii) The reaction is exothermic.  
Explain why, in terms of bonds broken and bonds formed.

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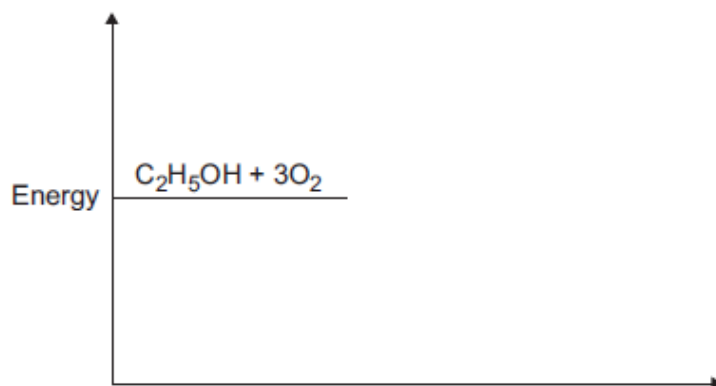
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(2)

(iii) Complete the energy level diagram for the combustion of ethanol.

On the completed diagram, label:

- activation energy
- overall energy change.



(3)

(Total 12 marks)

6.

One reason the oil industry is important is that it uses crude oil to produce many of the plastic materials we use in everyday life.

(a) The first stage in the formation of a plastic material is called cracking. Butane ( $C_4H_{10}$ ), a hydrocarbon in crude oil, can be cracked to produce two different hydrocarbons, ethane ( $C_2H_6$ ) and ethene ( $C_2H_4$ )

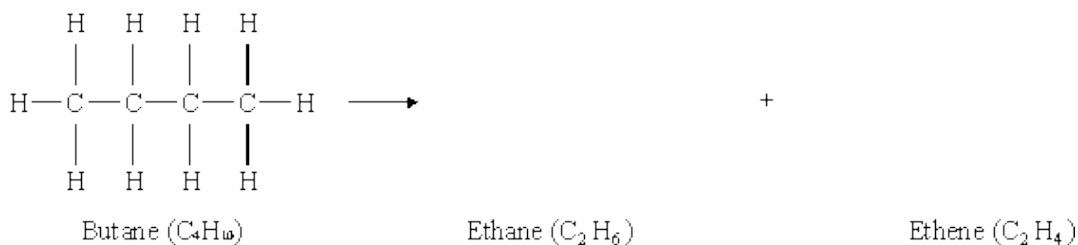
(i) For cracking to happen what needs to be done to the hydrocarbon?

\_\_\_\_\_

\_\_\_\_\_

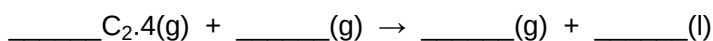
(2)

(ii) Complete the equation for the cracking of butane using displayed formulae.



(2)

(iii) Complete the balanced chemical equation for the complete combustion of ethane in oxygen.



(3)

- (b) The second stage is the formation of the plastic material by polymerisation.

Describe how ethene ( $C_2H_4$ ) forms poly(ethene). You do not need to give the reaction conditions or the names of catalysts.

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(3)

(Total 10 marks)

7.

Crude oil is a fossil fuel.

- (a) Describe how crude oil is separated into fractions.

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(4)

- (b) Fuel oil is one of the fractions from crude oil.

Power stations burn fuel oil to generate electricity. The waste gases from the combustion of fuel oil contain carbon dioxide, water vapour, sulfur dioxide and oxides of nitrogen.

The waste gases are passed through a suspension of limestone in water. Limestone is mainly calcium carbonate.

Suggest how the use of a suspension of limestone decreases one of the environmental impacts that the waste gases would cause.

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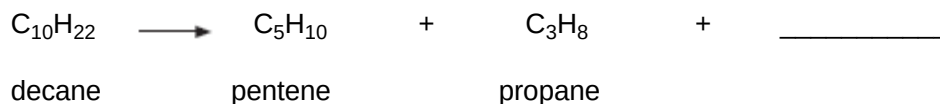
(3)

- (c) Some fractions from crude oil contain large hydrocarbon molecules.

- (i) Hydrocarbon molecules, such as decane, can be cracked to produce smaller, more useful molecules.

Write the correct formula of the third product to complete the chemical equation.

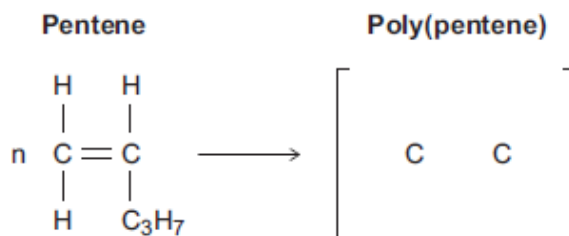
You do not need to give the name of this product.



(1)

- (ii) Pentene is used to produce poly(pentene).

Complete the equation and the displayed structure of poly(pentene).



(3)

(iii) Some polymers are described as smart polymers.

Suggest **one** property of a smart polymer that is different to that of an ordinary polymer.

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(1)

(Total 12 marks)



## Mark schemes

- 1.** (a) water  
*allow H<sub>2</sub>O* 1  
*allow hydrogen chloride or HCl* 1
- (b) single C–C bond and nothing added to the trailing bonds 1
- 3 × H and CH<sub>3</sub> correct  
*must be four single bonds* 1
- n at bottom right 1
- must be fully correct to score all 3 marks*  
*an answer of*
- $$n \begin{array}{c} \text{CH}_3 \quad \text{H} \\ | \quad | \\ \text{C} = \text{C} \\ | \quad | \\ \text{H} \quad \text{H} \end{array} \longrightarrow \left( \begin{array}{c} \text{CH}_3 \quad \text{H} \\ | \quad | \\ \text{C} - \text{C} \\ | \quad | \\ \text{H} \quad \text{H} \end{array} \right)_n$$
- scores 3 marks*
- (c) any **two** from:
- poly(propene) comes from a non-renewable source  
*allow poly(propene) will run out*
  - poly(propene) requires a lot of energy to make
  - poly(propene) is not biodegradable
  - a wool carpet needs replacing more often  
*must refer to the carpet, not just the fibre*
  - wool requires the use of large areas of land (which could be used to grow food crops)  
*ignore references to cost*  
*ignore pollution*  
*ignore landfill*  
*allow converse arguments*

2

(d) any **four** from:

advantages of polyester

- better flame resistance (so burns less easily)  
*allow good flame resistance so protects the firefighter*
- higher melting point (so melts less easily)  
*allow high melting point so uniform is not damaged*
- absorbs water so less likely to ignite

disadvantages of polyester:

- high density so uniform is heavy
- absorbs water so firefighter gets wet
- absorbs water so uniform becomes heavy
- justified conclusion

4

*allow converse arguments throughout.*

**max 3** marks if only advantages or only disadvantages  
of one type of fibre

[10]

2.

(a) wood is renewable  
**or**  
(natural) gas is finite

1

(burning) wood produces the same amount of carbon dioxide as the trees absorbed  
*allow wood is carbon-neutral allow wood does not add  
to global warming*

**or**

(burning natural) gas increases the amount of carbon dioxide (in the atmosphere)  
*allow (burning natural) gas adds to global warming  
allow (burning natural) gas adds greenhouse gases (to  
the atmosphere)  
ignore references to energy / cost*

1

(b) not enough oxygen  
*allow not enough air  
do **not** accept no oxygen / air*

1

(so) incomplete combustion

1

(c)  $2\text{CH}_4(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}(\text{g}) + 4\text{H}_2\text{O}(\text{g})$   
*allow correct multiples / fractions*

1

(d)

*an answer of 1250 (cm<sup>3</sup> oxygen unreacted) scores 4 marks*

ratio of O<sub>2</sub> : CO<sub>2</sub> = 5 : 3

1

$$\begin{aligned} \text{(oxygen needed)} &= \frac{3.60 \times 5}{3} \\ &= 6.0 \text{ (dm}^3\text{)} \end{aligned}$$

*allow correct calculation using an incorrectly determined mole ratio*

1

$$\text{(oxygen unreacted)} = 7.25 - 6.0 = 1.25 \text{ (dm}^3\text{)}$$

*allow correct subtraction of an incorrectly calculated volume of oxygen*

1

$$\text{(oxygen unreacted)} = 1.25 \times 1000$$

$$= 1250 \text{ (cm}^3\text{)}$$

*allow correct conversion to cm<sup>3</sup> anywhere in response*

1

**alternative approach for MP1 and MP2**

$$\text{moles CO}_2 = 0.15$$

**and**

$$\text{moles O}_2 = 0.25 \text{ (1)}$$

$$(0.25 \times 24 \Rightarrow) 6.0 \text{ (dm}^3 \text{ oxygen needed) (1)}$$

**[9]**

**3.**

any **four** from:

*to gain 4 marks both pros and cons should be given*

### **Arguments for biodiesel**

max **three** from:

- sustainable / renewable
- (carbon neutral) absorbs CO<sub>2</sub> when growing / during photosynthesis
- burning biodiesel produces low amounts particulates / carbon monoxide  
*allow burning biodiesel produces little / low amount of global dimming*  
*ignore sulfur dioxide*
- can use waste vegetable oils / fats (from food industry) **or** can use waste plant material
- can be used to conserve crude oil (instead of / mixed with petroleum diesel)
- produced by a low energy / temperature process  
*accept produced by a low tech process*
- biodegrades (easily)  
*ignore engine effects*

### **Arguments against biodiesel**

max **three** from:

- creates food shortages  
*accept price of food increases*
- deforestation to plant more crops leads to loss of habitat / biodiversity **or** deforestation leads to a reduction in absorption of CO<sub>2</sub>  
*allow burning trees increases CO<sub>2</sub>*  
*allow deforestation increases global warming*
- burning biodiesel produces high amounts of nitrogen oxides  
*allow increases acid rain*
- crops takes time to grow  
*allow crops can fail*
- vast areas of land needed to grow crops

4

conclusion supported by the argument presented, which must give added value to the points for and against given above

1

**[5]**

4.

- (a) (i) the products are at a lower energy level than the reactants  
*accept products have less energy / less energy at the end than the beginning*

1

- (ii) because a catalyst provides an alternative / different pathway / mechanism / reaction route

*accept adsorption or 'increases concentration at the surface'*  
*ignore absorption*

1

(that has) lower activation energy

*allow weakens bonds*

*allow idea of increased successful collisions.*

*DO NOT ALLOW answers stating catalysts provide energy for M1 and M2*

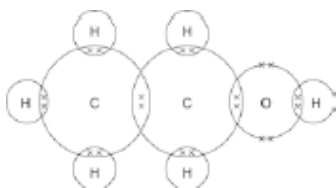
1

- (b) one pair of electrons in each overlap (8 pairs in total)

*allow any combination of dots, crosses or other symbols*

1

the rest of the diagram correct with four non-bonding electrons on the oxygen giving a total of eight electrons in oxygen outer energy level.



*gains 2 marks*

1

(c) (i)  $\pm 3024$  (J)

correct answer with or without working gains **3** marks

if the answer is incorrect, award up to **2** marks for the following steps:

- $\Delta T = 14.4(^{\circ}\text{C})$
- $50 \times 4.2 \times 14.4$

allow ecf for incorrect  $\Delta T$

3

(ii) 0.015(2173913)

correct answer with or without working gains **3** marks

if answer is incorrect, allow 1 mark each for any of the following steps up to a max of 2.

- 0.70g
- $M_r$  of ethanol = 46
- $0.70 / 46$

allow ecf in final answer for arithmetical errors

3

(iii)  $\pm 198\,720$  (J / mole)

$c(i) \div c(ii)$

allow ecf from **(c)(i)** and **(c)(ii)**

0.015 gives 201600

0.0152 gives 198947

0.01522 gives 198686

1

(d) (as the molecules get bigger **or** the number of carbon atoms increases) the intermolecular forces

allow intermolecular bonds

1

(intermolecular forces) increase

allow more / stronger (intermolecular forces)

1

and therefore require more (heat) energy to overcome

breaking covalent bonds or unspecified bonds max **1** mark (M3)

1

[15]

5.

(a)  $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

allow multiples

1

(b) 3444 J

if answer incorrect:

one mark for temperature increase =  $16.4^{\circ}\text{C}$

one mark for mass of water = 50 g

ecf for one incorrect value gains two marks for correct calculation

no ecf for two incorrect values

3

- (c) (i) 1276 (kJ per mole)

*ignore + or -*

*if answer incorrect:*

$$[(5 \times 413) + 347 + 358 + 467] + [(3 \times 495)] = 4722 \text{ (1 mark)}$$

$$[(4 \times 799) + (6 \times 467)] = 5998 \text{ (1 mark)}$$

*correct subtraction of calculated energy values (1 mark)*

3

- (ii) because energy released when bonds form is greater than energy used when bonds broken

*allow converse*

*if no mark awarded allow one mark for energy is used to break bonds*

*or*

*one mark for energy is released when bonds form*

2

- (iii) products line lower than reactants

1

activation energy labelled

1

overall energy change labelled

1

[12]

6.

- (a) (i) heat

*accept increase temperature ignore pressure*

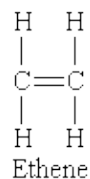
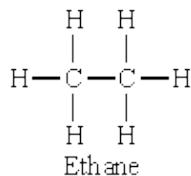
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with a catalyst

1

- (ii)

1

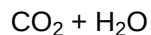


*accept displayed formulae only*

1

(iii) O<sub>2</sub>

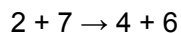
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*ignore state symbols*

1

correct balancing



*accept  $1 + 3\frac{1}{2} \rightarrow 2 + 3$  only if reactants and products correct*

1

(b) double bond breaks

1

**many** (ethene) molecules

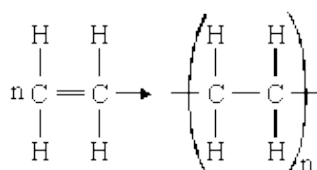
*accept many monomers*

1

bond together

*accept join **or** combine for bond*

*accept*



*for first 2 marks ignore unsaturated becomes saturated*

1

[10]

7.

(a) any **four** from:

- (crude oil is) heated
- to evaporate / vaporise / boil (the substances / hydrocarbons)
- the column is hotter at the bottom or is cooler at the top
- (vapours / fractions) condense
- at their boiling points or at different levels.

*marks can be taken from a diagram*

*max 3 marks for reference to cracking*

*allow fractional distillation allow vapours (enter the column)*

*allow temperature gradient or (vapours) cool as they rise*

*allow description e.g. vapour turns to liquid*

*allow they have different boiling points*

4



- (b) acid rain is caused by  
*allow consequences of acid rain*  
 1
- sulfur dioxide or oxides of nitrogen  
*second marking point is dependent on first marking point*  
 1
- they react with / are neutralised by calcium carbonate or limestone  
**OR**  
 global warming is caused by  
 carbon dioxide
- carbon dioxide will react or dissolve in suspension of limestone  
*allow greenhouse effect is caused by or allow consequences of global warming*  
 1
- (c) (i)  $C_2H_4$   
*must be formula*  
*ignore any name*  
 1
- (ii) a single bond between carbon atoms
- 
- would score 3 marks*  
 1
- other four bonds linking hydrogen atoms and  $C_3H_7$  group plus two trailing / connecting bonds  
 1
- n at the bottom right hand corner of the bracket  
 1
- (iii) has a shape memory  
**or**  
 (a smart polymer) can return to original shape (when conditions change)  
 1

[12]

## Examiner reports

1.

- (a) There was a variety of incorrect answers for the small molecule lost during condensation polymerisation, some of which were not molecules. However, 56% of students knew the answer was water.
- (b) Of those students who lost marks here, there was no clear-cut marking point that caused the problem. Some students did not make it clear that the bond joined on to the C atom of the methyl group, but were given benefit of the doubt provided it didn't join to the H3. 28% of students gave faultless answers.
- (c) This question was not answered well, mainly because students did not appreciate that they were making a comparison with both of the fibres used alone.
- Some students restated the question stem by saying 'wool wears out quickly'.
  - There was little mention of land usage for wool production or that poly(propene) is not biodegradable.
  - There were many uncreditworthy references to cost, and the greater comfort of wool.
  - Many students failed to make it clear whether they were talking about the fibre or the carpet, or whether they were comparing the mixed fibre carpet with wool or poly(propene) alone.

Most students who achieved any marks at all (39%) were only able to score one of the available marks.

- (d) This was generally well answered with 99% of students achieving at least one mark. There were various advantages and disadvantages given for the uniform absorbing water as covered by the marking points on the mark scheme (including an advantage of 'it would absorb lots of sweat'). 44% of the students were able to achieve all four marks.

2.

- (a) The majority of students could identify wood as a renewable resource or methane as a finite resource, however few could state that the burning of wood is carbon neutral. Some students however were able to state that wood does not contribute to global warming. While the vast majority scored one mark, fewer than 20% scored both.
- (b) This question was answered well, with around half of the students gaining both marks. Marginally more students knew the term 'incomplete combustion' than could explain its cause. Some students stated that generally there was never enough oxygen in air to combust methane completely, or that there was no oxygen available at all.
- (c) Balancing equations is a skill that is examined regularly but is still not performed well by some students. However, nearly four-fifths did score this mark. Correct examples of multiple/fractional balancing were seen.
- (d) This appeared to be an unfamiliar concept for most students and they found this question difficult. Some students could identify the 5:3 ratio from the equation, whilst others were able to convert from  $\text{dm}^3$  to  $\text{cm}^3$  and hence gain some marks. Many students did not appear to be able to bring the fact that the volume of any gas is proportional to the number of moles present to this question, and therefore involved the  $24 \text{ dm}^3$  molar gas volume in their calculation. Many others did no more than subtract  $3.60 \text{ dm}^3$  from  $7.25 \text{ dm}^3$  and convert the answer to  $\text{cm}^3$ . A common error was to divide by 100 instead of 1000 when converting from mg to g, or to multiply by 1000 instead. Fewer than 10% were able to score three or four marks, and more than half gained no mark at all.

3.

A surprising number of scripts were left blank. Good candidates produced cogent, well argued evaluations which scored a maximum of four marks, whereas weaker candidates were unable to do more than simply repeat the information given or use general or vague statements such as 'eco-friendly', 'cause pollution', 'exhaust fumes'. Specific examples were needed for candidates to gain credit.

Very few candidates provided a conclusion for one mark that added value to the rest of their answer. The vast majority were content to just summarise their previous points and then to declare themselves for or against the use of biodiesel. Good examples of conclusions that were seen made reference to the ethics of using land for cash crops rather than for food crops, to the possibilities of research into developing alternative fuel sources, and to the need for education in energy conservation.

Many candidates correctly interpreted the graph and correctly stated the consequences of particulates or acidic nitrogen oxide pollutants, although some confused carbon monoxide with carbon dioxide. However, many failed to notice that it was a percentage change so concluded incorrectly that no carbon dioxide was given off.

When discussing the possible disadvantages of using biodiesel, weaker candidates simply reiterated all information that was already given, such as, increased biodiesel use would lead to deforestation, or the use of crops for biodiesel that could be used for food. The stronger candidates gained marks by explaining exactly why deforestation was undesirable, and by referring to possible food shortages if vast areas of land were given over to the production of crops for biodiesel.

4.

(a) (i) Although most of the answers seen gained the mark, the most common error was to ignore the diagram and just give a definition of exothermic.

(ii) A small minority claimed that catalysts provided energy and so could not score any marks. Some students seemed to have cause and effect reversed; making statements which suggested that lowering the activation energy changed the route.

(b) The most common error was to miss out the non-bonding electrons on the oxygen. A few students either did not look at, or did not understand, the displayed structure of ethanol provided. Hence they either missed out bonding pairs or introduced double bonds.

(c) (i) Despite some careless errors (such as adding rather than multiplying or working out the temperature change incorrectly) most students gained all three marks available.

(ii) While many fully correct answers were seen, marks were often lost for excessive rounding (giving the number of moles as 0.02), carelessness (orders of magnitude changing or correct numbers giving incorrect answers). Students should be encouraged to check calculations through with care.

(iii) The most common error was to multiply the two previous answers rather than divide. The units given were " $\text{J / mol}$ " which should have helped students realise that joules need to be divided by moles.

(d) This was the most poorly answered question on the paper. Few students showed an understanding that intermolecular forces had to be overcome. The majority suggested that covalent bonds were broken. Some students who did mention intermolecular forces then contradicted themselves by either stating that more bonds were broken or stated the intermolecular forces were between atoms.

5.

(a) The majority of students were able to balance this equation. Multiples were occasionally seen and were given credit.

- (b) This question challenged many. The majority of students incorrectly substituted the mass of alcohol burned into the equation rather than the mass of water being heated. These students almost always scored two marks. A small number were unable to calculate the temperature increase correctly.
- (c) (i) This calculation also provided challenge but was quite well done. The sum of the bond energies on the left of the equation was harder to calculate correctly than that of those on the right, so students frequently gained two marks for a correct product side calculation and then an error carried forward correct subtraction.
- (ii) Poorly attempted – nearly half of the students scored no marks. Many students still believe that energy is needed both to break and form bonds. Others got the relative magnitude of reactant and product bond energies the wrong way around. A compensation mark was awarded for indicating either that energy is required to break bonds or that energy is released when bonds form.
- (iii) A very wide range of responses and a very even spread of marks as a result. Students who drew an endothermic diagram could and sometimes did score both the labelling marks. Other common errors were arrows pointing at the peak of the curve for activation energy, and arrows the full height from the peak down to the products line for overall energy change. The standard of presentation was frequently very poor, resulting in doubt as to whether credit should be awarded when arrows or lines showing energy changes stopped well short of target or overshoot.

## 6.

### Double and Single Award

In (a)(i) many candidates knew the conditions needed for cracking, although some omitted the use of a catalyst. Most were able to draw displayed formulae for ethane and ethene. The common error was to draw a single covalent bond between the carbon atoms of ethene. Part (iii) saw the correct completion of the chemical equation for the combustion of ethane although several candidates struggled to balance the equation. Most candidates knew that poly(ethene) was formed from ethene but many could not describe precisely what happens to the double covalent bonds in the many ethene molecules during polymerisation. The prevention of corrosion was generally missed, or the mark was not awarded on account of reference to aluminium rusting.

7.

- (a) The question was well answered with several students gaining full marks. Most students had a good appreciation of the key concepts of heating, evaporation and condensing, although the idea of the temperature gradient within the fractionating column was not seen so often. Sometimes the explanation of the vapours condensing at different boiling points was confusing and poorly explained as to how the fractions were collecting at different levels in the column. Students should use shorter, well-constructed sentences that include only one key point at a time.
- (b) The question was not well answered by large numbers of students. Although it was clear that many students could identify an environmental impact and the gas involved, the understanding of the removal of the gas was poorly explained. There was a split between students who answered the question by using acid rain with sulfur dioxide and global warming with carbon dioxide. Some students mentioned the greenhouse effect, although global warming, as an environmental impact, was a better answer. Oxides of nitrogen were rarely used to explain the impact of acid rain. The main problem was that many students confused the removal of these gases in industry with the test for carbon dioxide in the laboratory using limewater. This meant that these students were restricted to two marks for the environmental impact and gas involved.
- (c) (i) A large percentage of students gave the correct answer of  $C_2H_4$ , with many of them also including the name ethene. As expected, there was a wide range of incorrect answers, the most common being  $C_2H_6$ ,  $CH_4$  and  $C_{12}H_{24}$ . The last formula came from a lack of understanding of the equation, with students thinking that decane and pentene were the reactants producing propane and the unknown product.
- (ii) The vast majority of students gained some marks, with many getting full marks. The most common error was not including the trailing bonds or 'n' in the correct place outside the brackets. There were a few students who made the error of drawing a double bond between the two carbons.
- (iii) Many students had a good knowledge of smart polymers. While most correct answers involved a reference to shape memory polymers in a variety of ways, some students mentioned other uses of smart polymers. Often the description of the change of shape was not clear enough and could also have been applied to a thermosoftening polymer, such as, 'the polymer changes shape when heated'.